

Day 4: Loops in Python (Detailed Notes for Data Analysis Students)

What is a Loop?

A loop is used to execute a block of code repeatedly.

Instead of writing:

```
print("Welcome")  
print("Welcome")  
print("Welcome")  
print("Welcome")  
print("Welcome")
```

We can use:

```
for i in range(5):  
    print("Welcome")
```

Output:

```
Welcome  
Welcome  
Welcome  
Welcome  
Welcome
```

Why are Loops Important?

In Data Analysis, loops help:

- Process thousands of records

- Read multiple files
- Automate reports
- Analyze datasets
- Generate summaries

Example:

Imagine 10,000 employee records.

Without loops:

- Impossible to process manually

With loops:

- Easy and fast
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Types of Loops

1. for Loop
 2. while Loop
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1. FOR LOOP

Used when the number of repetitions is known.

Syntax

```
for variable in sequence:  
    statements
```

Example 1

```
for i in range(5):  
    print(i)
```

Output:

```
0  
1  
2  
3  
4
```

Understanding range()

range(5)

```
for i in range(5):  
    print(i)
```

Output:

```
0  
1  
2  
3  
4
```

range(1,6)

```
for i in range(1,6):  
    print(i)
```

Output:

```
1  
2  
3
```

```
4  
5
```

range(start, stop, step)

```
for i in range(1,11,2):  
    print(i)
```

Output:

```
1  
3  
5  
7  
9
```

Example 2: Print Name 5 Times

```
for i in range(5):  
    print("Umesh")
```

Example 3: Multiplication Table

```
num = 5  
  
for i in range(1,11):  
    print(num * i)
```

Output:

```
5  
10  
15  
20  
25
```

```
30
35
40
45
50
```

Example 4: Sum of Numbers

```
total = 0

for i in range(1,6):
    total += i

print(total)
```

Output:

```
15
```

Loop Through List

```
employees = ["Rahul","Amit","Priya"]

for emp in employees:
    print(emp)
```

Output:

```
Rahul
Amit
Priya
```

2. WHILE LOOP

Used when the number of repetitions is not known.

Syntax

```
while condition:  
    statements
```

Example

```
i = 1  
  
while i <= 5:  
    print(i)  
    i += 1
```

Output:

```
1  
2  
3  
4  
5
```

Infinite Loop

```
while True:  
    print("Hello")
```

⚠ This loop never stops.

Difference Between FOR and WHILE

FOR LOOP	WHILE LOOP
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Fixed repetitions	Unknown repetitions
Easier	More flexible
Uses range()	Uses condition

BREAK Statement

Used to stop the loop immediately.

Example

```
for i in range(1,11):  
  
    if i == 5:  
        break  
  
    print(i)
```

Output:

```
1  
2  
3  
4
```

CONTINUE Statement

Skips current iteration.

Example

```
for i in range(1,6):  
  
    if i == 3:
```

```
        continue

    print(i)
```

Output:

```
1
2
4
5
```

PASS Statement

Used as a placeholder.

```
for i in range(5):
    pass
```

No output.

Nested Loops

Loop inside another loop.

Example

```
for row in range(1,4):

    for col in range(1,4):

        print("*", end=" ")

    print()
```

Output:


```
* * *  
* * *  
* * *
```

Pattern Programs

Pattern 1

```
for i in range(5):  
  
    print("*")
```

Output:

```
*  
*  
*  
*  
*
```

Pattern 2

```
for i in range(1,6):  
  
    print("*" * i)
```

Output:

```
*  
**  
***  
****  
*****
```

Real Data Analyst Examples

Example 1: Total Sales

```
sales = [10000,20000,30000,40000]

total = 0

for s in sales:
    total += s

print("Total Sales =", total)
```

Output:

```
Total Sales = 100000
```

Example 2: Find High Sales

```
sales = [15000,45000,60000,25000]

for s in sales:

    if s > 30000:
        print(s)
```

Output:

```
45000
60000
```

Example 3: Employee Salary Increment

```
salary = [25000,30000,35000]

for s in salary:
```

```
new_salary = s * 1.10

print(new_salary)
```

Output:

```
27500
33000
38500
```

Mini Project 1: Multiplication Table

```
num = int(input("Enter Number: "))

for i in range(1,11):

    print(num,"x",i,"=",num*i)
```

Mini Project 2: Sum of First N Numbers

```
n = int(input("Enter Number: "))

total = 0

for i in range(1,n+1):

    total += i

print(total)
```

Mini Project 3: Factorial

```
n = int(input("Enter Number: "))
```

```
fact = 1

for i in range(1,n+1):

    fact *= i

print(fact)
```

Mini Project 4: Employee Salary Report

```
salary = [25000,35000,45000]

for s in salary:

    print("Salary =",s)
```

Practice Questions

Basic Loops

1. Print numbers 1 to 10.
 2. Print numbers 1 to 20.
 3. Print even numbers.
 4. Print odd numbers.
 5. Print your name 10 times.
-

FOR Loop

6. Print table of 2.
7. Print table of 10.
8. Find sum of 1–100.
9. Find square of numbers.
10. Find cube of numbers.

WHILE Loop

- 11. Print 1–10.
- 12. Print 10–1.
- 13. Print even numbers.
- 14. Print odd numbers.
- 15. Print table of a number.

BREAK

- 16. Stop at 5.
- 17. Stop at 10.
- 18. Stop when number found.
- 19. Exit loop after condition.
- 20. Create search system.

CONTINUE

- 21. Skip number 5.
- 22. Skip even numbers.
- 23. Skip odd numbers.
- 24. Skip multiples of 3.
- 25. Skip negative values.

Nested Loops

- 26. Print square pattern.
- 27. Print triangle pattern.
- 28. Print multiplication grid.
- 29. Print employee report matrix.
- 30. Print star patterns.

Business Problems

31. Calculate total sales.
 32. Calculate average sales.
 33. Count employees.
 34. Find highest salary.
 35. Find lowest salary.
 36. Count products sold.
 37. Generate invoice numbers.
 38. Display monthly sales.
 39. Calculate yearly sales.
 40. Create salary increment report.
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Assignment (Data Analyst Level)

Create a Sales Report Program

```
sales = [12000,25000,35000,45000,55000]
```

Display:

```
Month 1 : 12000
Month 2 : 25000
Month 3 : 35000
Month 4 : 45000
Month 5 : 55000
```

```
Total Sales : ?
Average Sales : ?
Highest Sales : ?
Lowest Sales : ?
```

Interview Questions

1. What is a loop?
2. Why are loops used?
3. Difference between for and while?
4. What is range()?
5. What is break?
6. What is continue?
7. What is pass?
8. What is nested loop?
9. How are loops used in Data Analysis?
10. What is an infinite loop?